

11. Upravljanje tuljara kondenzator u tobovima i izmjenici tobova

Estetiziraj komo 8 nerzi RLC u slepti, meriti komo furi komil signola to dozirano tulor, da dolozio kapaciteta induktivit upom in fukure vira

$$I_0 = \sqrt{I_R^2 + (I_C - I_L)^2}$$

$$\text{kap. upom } X_C = \frac{1}{\omega C}$$

$$\text{ind. upom } X_L = \omega L$$

$$\Rightarrow \varphi = \arctan \frac{I_C - I_L}{I_R}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

1. nerava roporedna

$$I_0 = 1,6 \text{ mA}$$

$$\text{upom: } R = 2700 \Omega$$

$$U_C = 2,1 \text{ V}$$

redica njolira njova iuna

$$U_L = 0,1 \text{ V}$$

$$\text{tuljara: } L = (0,25 \pm 0,05) \text{ H} \leftarrow \text{pol one tipove}$$

$$U_R = 0,21 \text{ V}$$

$$\text{kondenzator: } C = 2 \mu\text{F} \pm 0,1 \mu\text{F}$$

1200
mouly

$$U_0 = 2,3 \text{ V}$$

4. nerava: CL bes R

2. nerava CR

$$I_0 = 2,5 \text{ mA}$$

$$I_0 = 2,0 \text{ mA}$$

$$U_C = 3,8 \text{ V}$$

$$U_0 = 2,5 \text{ V}$$

$$U_R = 0,38 \text{ V} \leftarrow \text{meno 1}$$

$$U_C = 2,5 \text{ V}$$

$$U_L = 0,1 \text{ V}$$

$$U_0 = 3,8 \text{ V}$$

7
vrepredhe

3. nerava L, R

$$I_0 = 4,2 \text{ mA}$$

$$U_R = 1,1 \text{ V}$$

$$U_L = 0,03 \text{ V}$$

$$U_0 = 1,1 \text{ V}$$

vezova 5. ve RLC napredne

$$U_0 = 7,7$$

$$I_R = 28 \text{ mA}$$

$$I_C = 5 \text{ mA}$$

$$I_L = 0,38 \text{ A}$$

vezova 6. RL

$$U_0 = 4,6 \text{ V}$$

$$I_R = 17 \text{ mA}$$

$$I_L = 0,22 \text{ A}$$

vezova 7. CR

$$U_0 = 4,0 \text{ V}$$

$$I_R = 15 \text{ mA}$$

$$I_C = 2,6 \text{ mA}$$

vezova 8. CL

$$U_0 = 5,1 \text{ V}$$

$$I_C = 3,4 \text{ mA}$$

$$I_L = 250 \text{ mA}$$

(A) (b)

ПОРОЇЛО 11: RLC мережа в тоболовж

1. RLC мережа. $\omega = 2\pi f = 314 \frac{1}{s}$ $\delta R = 2\%$ $\delta L = 5\%$ $\delta C = 5\%$

$$Z_T = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{(270\Omega)^2 + \left(314 \frac{1}{s} \cdot 0,25 \text{ H} - \frac{1}{314 \frac{1}{s} \cdot 2 \cdot 10^{-6} \text{ F}}\right)^2}$$

$$Z_T = (1500 \pm 100) \Omega$$

$$\phi_T = \arctan \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right) = \dots = -88^\circ \pm 5^\circ$$

$$R = (270 \pm 10) \Omega$$

$$L = (0,25 \pm 0,05) \text{ H}$$

$$C = 2 \mu\text{F} \pm 0,1 \mu\text{F}$$

$$Z_i = \frac{U_0}{I_0} = \frac{2,3 \text{ V}}{0,0016 \text{ A}} = (1400 \pm 100) \Omega$$

$$\phi_i = \arctan \left(\frac{U_L - U_C}{U_R} \right) = \arctan \left(\frac{\overset{10\%}{0,01 \text{ V}} - \overset{2\%}{2,1 \text{ V}}}{\underset{3\%}{0,21 \text{ V}}} \right) \approx \overset{\sim 5\%}{-84^\circ \pm 4^\circ}$$

2. RC мережа.

$$Z_T = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2} = (130 \pm 10) \Omega$$

$$\phi_T = \arctan \left(\frac{-\frac{1}{\omega R}}{R} \right) = -80^\circ \pm 4^\circ$$

$$Z_i = \frac{U_0}{I_0} = \frac{3,8 \text{ V}}{0,0025 \text{ A}} = (150 \pm 10) \Omega$$

$$\phi_i = \arctan \left(-\frac{U_C}{U_R} \right) = -84^\circ \pm 4^\circ$$

3. LR мережа.

$$Z_T = \sqrt{R^2 + (L\omega)^2} = 280 \Omega \pm 20 \Omega$$

$$\phi_T = \arctan \left(\frac{\omega L}{R} \right) = 89^\circ \pm 5^\circ$$

$$Z_i = \frac{U_0}{I_0} = 260 \Omega \pm 20 \Omega$$

$$\phi_i = \arctan \left(\frac{U_L}{U_R} \right) = 1^\circ \pm 5^\circ$$

4. C.L. ~~nap~~

$$Z_T = \sqrt{\left(\omega L - \frac{1}{\omega C}\right)^2} = 9987 \Omega \pm 100 \Omega$$

$$Z_i = \frac{U_0}{I_0} = \frac{2,5V}{0,002A} = 1250 \Omega \pm 1000$$

$$\phi_T = ? \quad \text{deljeje } 20^\circ ?$$

5. RLC ~~napredno~~ ~~napredno~~ ~~napredno~~!

$$Z_T = \left(\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2} \right)^{-1} = (90^\circ \pm 5^\circ) \Omega$$

$$I_0 = \sqrt{I_R^2 + (I_C - I_L)^2} = 0,37A \quad Z_i = \frac{7,7V}{0,37A} = (20^\circ \pm 2^\circ) \Omega$$

$$\phi_T = \arctan \left(R \left(\omega C - \frac{1}{\omega L} \right) \right) = 73^\circ \pm 5^\circ$$

$$\phi_i = \arctan \left(\frac{I_C - I_L}{I_R} \right) = -85^\circ \pm 5^\circ$$

6. RL ~~napredno~~

$$Z_T = \left(\sqrt{\frac{1}{R^2} + \left(\frac{1}{\omega L}\right)^2} \right)^{-1} = 75 \Omega \pm 4 \Omega$$

$$I_0 = \sqrt{I_R^2 + I_L^2} = 0,22A$$

$$Z_i = \frac{U_0}{I_0} = \frac{4,6V}{0,22A} = 21 \Omega \pm 2 \Omega$$

$$\phi_T = \arctan \left(R \left(\omega - \frac{1}{\omega L} \right) \right) = -73^\circ \pm 5^\circ$$

$$\phi_i = \arctan \left(\frac{-I_L}{I_R} \right) = -85^\circ \pm 6^\circ$$

7. CR vzporedna

$$Z_T = \left(\sqrt{\frac{1}{R^2} + (\omega C)^2} \right)^{-1} = (266 \pm 4) \Omega$$

$$I_0 = \sqrt{I_R^2 + I_C^2} = 0,015 \text{ A} \quad Z_i = \frac{U_i}{I_0} = \frac{4,0 \text{ V}}{0,015 \text{ A}} = (266 \pm 4) \Omega$$

$$\phi_T = \arctan(R(\omega C)) = 10^\circ \pm 5^\circ$$

$$\phi_i = \arctan\left(\frac{I_C}{I_R}\right) = 10^\circ \pm 5^\circ$$

8. CL vzporedna

$$Z_T = \left(\sqrt{(\omega L - \frac{1}{\omega C})^2} \right)^{-1} = 83 \Omega \pm 4 \Omega$$

$$I_0 = |I_C + I_L| = 0,253 \text{ A} \quad Z_i = \frac{5,1 \text{ V}}{0,253 \text{ A}} = (20 \pm 4) \Omega$$

$$\phi = 0$$

g₁: identiteta $U_L = -\frac{d\Phi_B}{dt} = -L \frac{dI}{dt}$, kapacitivnost $C = \frac{Q}{U}$
merili bi $U = R \cdot I$ tužje in izračunali R.

g₂: farin ravnal v + smer tužje v ločen obkrožje

g₃: saj je miteje dejstva sinusoida, kože si je celoten predstaviti
v kompleksni ravnini

$$g_4: \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50 \cdot 10^{-9} \text{ F}}} = 124 \frac{1}{\text{s}} \Rightarrow \nu = 19 \text{ Hz}$$

$$\frac{dZ}{d\omega} = 0 \quad \frac{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}{d\omega} = 0 \dots$$